

Tutorial

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Redo of assignment 1

Question 2

0.0.1 (d)

$$|2, 1\rangle = \frac{1}{3}|\frac{3}{2}, \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\rangle + \sqrt{\frac{3}{4}}|\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle \quad (1)$$

Wrong $\sqrt{\frac{3}{4}}$ isn't the probability

$$|J, M_J\rangle = \sum_{m=0}^{\infty} C^{J_0, m_0, J_p, m_p} |j_0, m_0; j_0, m_0\rangle \quad (2)$$

I need to expand in coupled basis

$$|\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle = a|2, 1\rangle + b|1, 1\rangle \quad (3)$$

P total of 2 = $|a|^2$

$$|2, 1\rangle = \frac{1}{2}|\frac{3}{2}, \frac{3}{2}, \frac{1}{2}, \frac{1}{2}\rangle + \sqrt{\frac{3}{4}}|\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\rangle \quad (4)$$